Progress Report: MeanFlow Under Compressed and Bidirectional Transport

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1 Objectives

The main objectives of these experiments were:

- To study how MeanFlow behaves when one endpoint of a paired transport problem becomes highly compressed.
- To check whether the compressed mapping can still be reversed correctly.
- To compare one-step and few-step MeanFlow with ordinary Flow Matching and Rectified Flow.
- To determine whether poor reconstruction comes from information loss, limited neural-network capacity, insufficient training, or the MeanFlow training formulation.
- To use the findings as a controlled starting point for the later image-to-mask and mask-to-image thesis work.

2 Initial Research Motivation

Our broader thesis idea is related to bidirectional transport between an image and a segmentation mask:

$$\text{Image} \leftrightarrow \text{Mask}.$$

The difficulty is that an image contains much more information than a mask. Different images may produce the same segmentation mask.

For example,

$$I_1 \to M,$$

$$I_2 \to M,$$

$$I_3 \to M.$$

Therefore, the mask may not contain enough information to reconstruct the exact original image.

Before working directly with medical images, we decided to study a much simpler two-dimensional problem where the amount of endpoint compression could be controlled exactly.

3 Important Correction to the Original Hypothesis

At first, we thought MeanFlow might fail because it learns an "average velocity."

However, this interpretation was incorrect.

MeanFlow learns the **average velocity over a time interval**. It does not automatically average different training images or different samples together.

Therefore, we should not say:

> MeanFlow averages different samples and therefore produces blur.

Instead, our experimental question became:

> **Does MeanFlow become difficult to learn or reverse when one endpoint becomes extremely compressed, even though the mathematical inverse still exists?**

This became the main question of the initial toy experiments.

4 Toy Dataset

We generated four two-dimensional Gaussian clusters.

The original point is represented by

$$x_0.$$

For every cluster, the center is

$$c.$$

A compressed version of every point was created using

$$\boxed{x_1 = c + s(x_0 - c)}$$

where s is the compression factor.

The meaning of different values of s is shown in Table 1.

Table 1: Meaning of different compression factors.

Compression s	**Meaning**	**Approx. Reverse Expansion**
1.00	No compression	$1\times$
0.10	10% spread remains	$10\times$
0.05	5% spread remains	$20\times$
0.02	2% spread remains	$50\times$
0.01	1% spread remains	$100\times$

For every

$$s > 0,$$

the mapping is still mathematically reversible:

$$\boxed{x_0 = c + \frac{x_1 - c}{s}}$$

Therefore, $s = 0.01$ is very difficult, but it does not mathematically destroy the information.

Only when

$$s = 0$$

do all points inside one cluster collapse exactly to the cluster center.

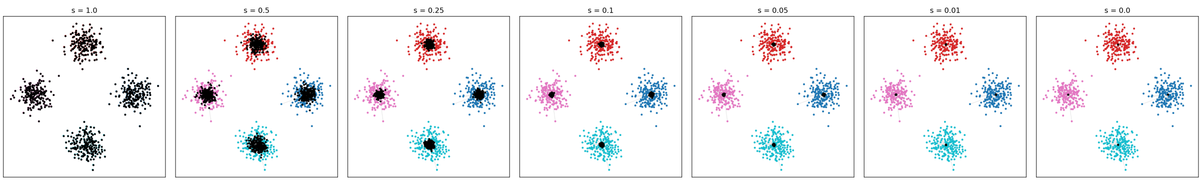


Figure 1: Effect of the compression factor s on the toy dataset. The colored points represent the original samples x_0, while the black points represent the compressed endpoints x_1. As s decreases, the within-cluster spread of x_1 becomes smaller. At $s = 0$, all compressed samples collapse to their corresponding cluster centers.

Figure 1 shows the effect of changing the compression factor. As s decreases, the compressed endpoint becomes increasingly thin.

5 Path Used for Flow Training

For both Flow Matching / Rectified Flow and MeanFlow, we use the same straight interpolation path:

$$\boxed{z_t = (1 - t)x_0 + tx_1}$$

where

$$0 \leq t \leq 1.$$

At

$$t = 0,$$

we obtain

$$z_0 = x_0.$$

At

$$t = 1,$$

we obtain

$$z_1 = x_1.$$

Therefore, this equation describes a straight path between the original point and the compressed point.

The true instantaneous velocity of this path is

$$\boxed{v = x_1 - x_0.}$$

Using the same path for both methods is important because it allows us to compare their learning mechanisms on the same transport problem.

6 Flow Matching and Rectified Flow

Rectified Flow learns the instantaneous velocity

$$v_\theta(z_t, t),$$

where:

- z_t is the current point,
- t is the current time,
- θ represents the neural-network parameters.

The correct training target for the straight path is

$$v = x_1 - x_0.$$

Therefore, the training loss is

$$\boxed{L_{\mathrm{RF}} = \|v_\theta(z_t, t) - v\|^2 .}$$

During forward sampling, Euler integration is used:

$$\boxed{z_{t+\Delta t} = z_t + \Delta t\, v_\theta(z_t, t).}$$

For backward sampling,

$$\boxed{z_{t-\Delta t} = z_t - \Delta t\, v_\theta(z_t, t).}$$

In the final comparison, we tested:

$$\text{RF-1}, \qquad \text{RF-4}, \qquad \text{RF-25}.$$

The number represents the number of neural-network evaluations used during sampling.

7 MeanFlow

MeanFlow uses a different idea.

Instead of learning only the instantaneous velocity, MeanFlow learns an average velocity over a time interval:

$$\boxed{u(z_t, r, t) = \frac{1}{t - r} \int_r^t v(z_\tau, \tau)\, d\tau.}$$

Here:

- z_t is the current state,
- t is the later time,
- r is the earlier time,
- u is the average velocity over the interval $[r, t]$,
- v is the instantaneous velocity.

The MeanFlow sampling equation is

$$\boxed{z_r = z_t - (t - r)u_\theta(z_t, r, t).}$$

Therefore, MeanFlow can potentially move across a large interval using only one neural-network evaluation.

For example, for one-step sampling,

$$t = 1, \qquad r = 0.$$

Then

$$z_0 = z_1 - u_\theta(z_1, 0, 1).$$

This is the main reason MeanFlow is interesting for fast one-step or few-step inference.

8 MeanFlow Training Identity

The MeanFlow training target is

$$\boxed{u_{\text{target}} = v - (t - r)\frac{du}{dt}.}$$

The total derivative is

$$\boxed{\frac{du}{dt} = v\frac{\partial u}{\partial z} + \frac{\partial u}{\partial t}.}$$

The first part,

$$v\frac{\partial u}{\partial z},$$

represents the change in u because the position z changes.

The second part,

$$\frac{\partial u}{\partial t},$$

represents the direct change in u because time changes.

In our implementation, this total derivative is calculated using a Jacobian-vector product (JVP).

The MeanFlow training loss is

$$\boxed{L_{\text{MF}} = \left\| u_\theta - u_{\text{target}} \right\|^2.}$$

The target is detached before the MSE loss is calculated.

9 Evaluation Metrics

We mainly used three evaluation metrics.

9.1 Reconstruction MSE

The reconstruction mean squared error is

$$\boxed{\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N} \left\| \hat{x}_{0,i} - x_{0,i} \right\|^2 .}$$

A smaller value is better.

This metric measures how close the reconstructed point is to the correct paired point.

9.2 Spread Ratio

The spread ratio is

$$\boxed{\text{Spread Ratio} = \frac{\text{Predicted Cluster Spread}}{\text{True Cluster Spread}}.}$$

A spread ratio close to 1 means the predicted cluster has approximately the correct size.

A value below 1 means the reconstruction is too compressed.

A value above 1 means the reconstruction is too expanded.

9.3 Alignment

For every true point, we define

$$d = x_0 - c.$$

For the predicted point,

$$\hat{d} = \hat{x}_0 - c.$$

The alignment is

$$\boxed{\text{Alignment} = \frac{d \cdot \hat{d}}{\|d\| \|\hat{d}\|}.}$$

An alignment close to 1 means that the model reconstructs the point in the correct sample-specific direction.

10 Experiment 1: Initial Flow Matching Baseline

We first tested ordinary Flow Matching at

$$s = 0.10.$$

The results are shown in Table 2.

Table 2: Initial Flow Matching results at $s = 0.10$.

Method	MSE	Spread Ratio
FM-1	0.0103	0.841
FM-50	0.0041	0.980

FM-50 reconstructed the distribution very accurately.

Therefore, the toy dataset and the basic Flow Matching implementation were working correctly.

11 Experiment 2: Initial MeanFlow Test

At the same compression,

$$s = 0.10,$$

MeanFlow produced the results shown in Table 3.

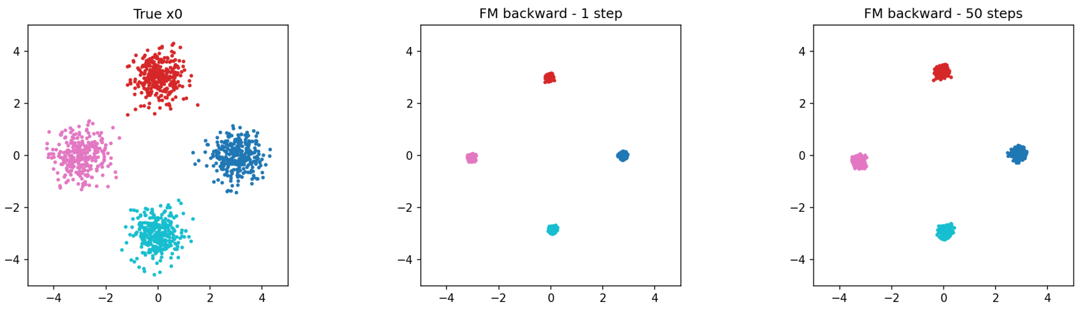
Table 3: Initial MeanFlow results at $s = 0.10$.

Method	MSE	Spread Ratio
MF-1	0.0077	1.026
MF-4	0.0020	0.992

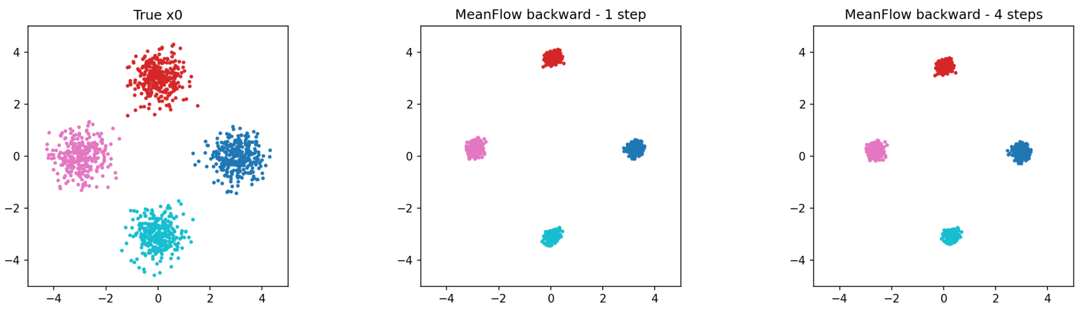
MeanFlow worked very well.

Therefore, our original idea that one-step MeanFlow would immediately fail when the endpoint became compressed was not supported.

Figure 2 gives a visual example of the backward reconstruction problem. The reconstructed cluster centers are generally recovered, but the amount of within-cluster spread recovered by the model depends on the method and number of sampling steps.



(a) Flow Matching backward reconstruction.



(b) MeanFlow backward reconstruction.

Figure 2: Qualitative examples of backward reconstruction during the initial experiments. The Flow Matching result compares the true x_0 distribution with one-step and 50-step reconstruction. The MeanFlow result compares the true distribution with one-step and four-step reconstruction. The figures show how the methods recover the locations and within-cluster spread of the original samples.

12 Experiment 3: Increasing the Compression

We then increased the compression to

$$s = 0.02.$$

The results are shown in Table 4.

Table 4: Results at $s = 0.02$.

Method	MSE	Spread	Alignment
FM-1	0.160	0.330	0.606
FM-50	0.105	0.483	0.735
MF-1	0.131	0.701	0.591
MF-4	0.115	0.591	0.603

MeanFlow produced a wider output distribution, but this wider distribution did not mean that it reconstructed the correct paired points.

For example, FM-50 had a lower spread ratio but better alignment.

Therefore, spread alone is not enough to judge successful inverse reconstruction.

13 Experiment 4: Extreme Compression

We next tested

$$s = 0.01,$$

which requires approximately

$$100\times$$

reverse expansion.

The early results are shown in Table 5.

Table 5: Early results at $s = 0.01$.

Method	MSE	Spread	Alignment
FM-1	0.211	0.245	0.375
FM-50	0.183	0.391	0.464
MF-1	0.246	0.714	0.374
MF-4	0.197	0.568	0.352

Again, MeanFlow produced a relatively large spread, but the sample-wise alignment remained poor.

Therefore, a large output spread does not necessarily mean that the original paired point was reconstructed correctly.

14 Experiment 5: Five-Seed Reproducibility Test

To make sure the result was not caused by one random neural-network initialization, we repeated the experiment using five seeds:

$$\{1, 2, 3, 4, 5\}.$$

The MF-1 alignment trend is shown in Table 6.

Table 6: Mean MF-1 alignment across five seeds.

Compression s	**MF-1 Alignment**
0.10	0.970
0.05	0.880
0.02	0.474
0.01	0.189

As compression became more extreme, sample-wise reconstruction became much worse.

The five-seed experiment showed that this degradation was reproducible and was not only caused by one unlucky random seed.

15 Experiment 6: MeanFlow Training-Length Test

We then asked:

> Is MeanFlow simply undertrained?

At

$$s = 0.01,$$

we increased the training length.

The results are shown in Table 7.

Going from 3000 to 6000 steps gave a large improvement.

However, going from 6000 to 12000 steps did not improve the alignment further:

$$0.354 \to 0.336.$$

Therefore, insufficient training was part of the problem, but simply training longer did not completely solve it.

Table 7: MF-1 performance with different training lengths.

Training	MSE	Spread	Alignment
3000 steps	0.367	0.923	0.194
6000 steps	0.215	0.663	0.354
12000 steps	0.196	0.527	0.336

16 Experiment 7: Direct MLP Control

We then performed an important control experiment.

Instead of using Flow Matching or MeanFlow, a normal multilayer perceptron (MLP) directly learned

$$\boxed{x_1 \to x_0.}$$

The purpose was to check whether an ordinary neural network could learn the difficult $100\times$ inverse.

16.1 First Direct MLP Test

With approximately 4,000 training points and 6,000 training steps, the results were:

Table 8: Initial direct MLP result.

Metric	Result
MSE	0.162
Spread	0.412
Alignment	0.552

The result was better than some MeanFlow results, but it was still not very accurate.

This suggested that limited training data might also be affecting the experiment.

16.2 Larger Direct MLP Test

We increased the training data to approximately 20,000 points and trained for 12,000 steps.

The result became:

This was an important result.

It showed that the $100\times$ inverse is learnable by a normal neural network when enough training data and training steps are provided.

Therefore, the compressed endpoint still contains enough information to reconstruct the original sample.

Table 9: Direct MLP result with more data and longer training.

Metric	Result
MSE	**0.023**
Spread	**0.867**
Alignment	**0.929**

17 Experiment 8: Matched MeanFlow Test

To make the comparison fairer, MeanFlow was also trained using approximately 20,000 training points and 12,000 training steps.

The MF-1 average result was:

Table 10: Matched MeanFlow result with 20,000 training points.

Metric	MeanFlow Result
MSE	0.276
Spread	0.751
Alignment	0.231
Tail Training Loss	0.655

For comparison, the direct MLP obtained

$$\text{MSE} = 0.023$$

and

$$\text{Alignment} = 0.929.$$

Therefore, the underlying inverse was clearly learnable, but our current MeanFlow training setup still had difficulty learning the same sample-wise reconstruction.

However, this experiment does not prove that MeanFlow fundamentally cannot perform the task.

Direct regression and MeanFlow use different training objectives.

The experiment only shows that, under our tested setup, MeanFlow optimization was much more difficult.

18 Experiment 9: Rectified Flow vs Shared Bidirectional MeanFlow

We then moved closer to the actual thesis problem.

Instead of studying only backward MeanFlow, we trained models for both directions:

$$x_0 \to x_1$$

and

$$x_1 \to x_0.$$

The Rectified Flow model learned an instantaneous velocity field.

The shared MeanFlow model used

$$u_\theta(z, r, t, d),$$

where d is a direction indicator.

The experiment used:

- hidden size = 128,
- 6,000 training steps,
- three random seeds,
- approximately 4,000 training samples,
- approximately 1,200 test samples.

We tested

$$s \in \{0.10, 0.05, 0.02, 0.01\}.$$

For Rectified Flow, we tested:

$$\text{RF-1}, \qquad \text{RF-4}, \qquad \text{RF-25}.$$

For MeanFlow, we tested:

$$\text{MF-1}, \qquad \text{MF-4}.$$

19 Main Rectified Flow vs MeanFlow Result

The clearest comparison is the difficult inverse direction:

$$x_1 \to x_0.$$

Table 11 compares RF-25 with MF-1.

At moderate compression,

Table 11: Backward reconstruction comparison between RF-25 and MF-1.

s	**RF-25 MSE**	**MF-1 MSE**	**RF-25 Align.**	**MF-1 Align.**
0.10	0.0018	0.0051	0.986	0.972
0.05	0.0119	0.0197	0.942	0.918
0.02	**0.0529**	**0.1307**	**0.838**	**0.594**
0.01	**0.1533**	**0.2976**	**0.503**	**0.271**

$$s = 0.10$$

and

$$s = 0.05,$$

both methods worked well.

However, at stronger compression,

$$s = 0.02$$

and

$$s = 0.01,$$

MeanFlow degraded more quickly than the multi-step Rectified Flow baseline.

For example, at

$$s = 0.02,$$

RF-25 alignment was

$$0.838,$$

while MF-1 alignment was

$$0.594.$$

At

$$s = 0.01,$$

RF-25 alignment was

$$0.503,$$

while MF-1 alignment decreased to

$$0.271.$$

Therefore, both methods became worse when compression increased, but the tested shared MeanFlow model degraded more strongly.

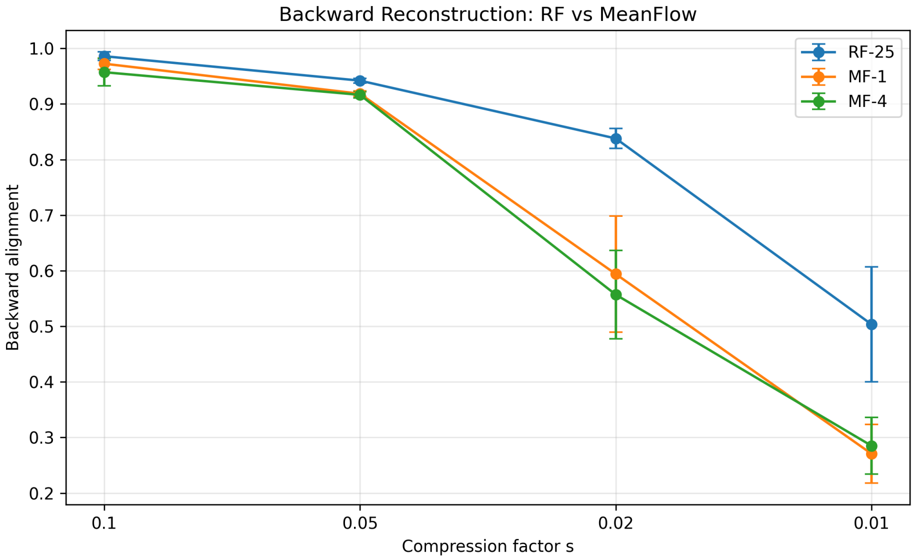


Figure 3: Backward reconstruction alignment for Rectified Flow and MeanFlow under increasing endpoint compression. RF-25, MF-1 and MF-4 perform similarly at moderate compression. However, a larger difference appears when s decreases to 0.02 and 0.01. Error bars show the standard deviation across three random seeds. Higher alignment is better.

As shown in Figure 3, RF-25 and MeanFlow perform similarly when compression is moderate.

However, a clear gap appears at

$$s = 0.02$$

and becomes larger at

$$s = 0.01.$$

This shows that the tested MeanFlow model becomes less accurate than the multi-step Rectified Flow baseline as the inverse transport becomes more difficult.

20 MeanFlow Training Loss During Compression

The MeanFlow tail training loss also increased as the endpoint became thinner.

This gives the general trend

Table 12: MeanFlow training loss under different compression levels.

Compression s	**MeanFlow Tail Loss**
0.10	0.009
0.05	0.051
0.02	0.239
0.01	0.343

$$\boxed{\text{More Compression} \rightarrow \text{Harder MeanFlow Optimization.}}$$

This observation is specific to the current controlled experimental setup.

21 One-Step vs Four-Step MeanFlow

We also checked whether increasing the number of MeanFlow steps could solve the difficult inverse.

At

$$s = 0.01,$$

the backward alignments were approximately

$$\text{MF-1} = 0.271$$

and

$$\text{MF-4} = 0.285.$$

The improvement was very small.

At

$$s = 0.02,$$

MF-4 was also not clearly better than MF-1.

Therefore, the current results do not suggest that the difficulty is caused only by taking one very large MeanFlow step.

22 Important Note About the Forward Direction

The forward direction is

$$x_0 \to x_1.$$

When

$$s = 0.01,$$

the true x_1 points are extremely close to their cluster centers.

Because the true spread becomes very small, the spread ratio and alignment can become numerically sensitive.

Therefore, for the strongly compressed forward direction, MSE is a more reliable measure.

For example, at

$$s = 0.01,$$

RF-25 achieved approximately

$$\text{MSE} = 0.000051,$$

while MF-4 achieved approximately

$$\text{MSE} = 0.00348.$$

Therefore, Rectified Flow also produced a more accurate compressed endpoint in this experiment.

23 Main Findings

The experiments currently support the following findings:

1. **MeanFlow does not automatically fail under endpoint compression.**

   At moderate compression such as

   $$s = 0.10$$

   and

   $$s = 0.05,$$

   MeanFlow performs very well.

2. **Extreme compression makes inverse reconstruction difficult for all tested flow methods.**

   Both Rectified Flow and MeanFlow became less accurate as s decreased.

3. **The direct MLP experiment showed that the $s = 0.01$ inverse is still learnable.**

   With sufficient training data, direct regression achieved approximately

   $$\text{Alignment} = 0.929.$$

   Therefore, poor reconstruction cannot simply be explained by complete information loss.

4. **Our current MeanFlow implementation did not learn the same inverse nearly as accurately under the tested conditions.**

5. **In the final bidirectional comparison, Rectified Flow degraded more slowly than the shared MeanFlow model as compression became severe.**

6. **MeanFlow training loss increased strongly as compression increased.**

   This suggests that MeanFlow optimization became more difficult in this controlled setting.

7. **Using four MeanFlow steps did not completely solve the difficult inverse.**

   Therefore, the observed difficulty does not appear to be caused only by using one large interval jump.

24 What We Are Not Claiming

At this stage, we should not claim that MeanFlow fundamentally fails for compressed distributions.

We should also not claim that the JVP is definitely responsible for the observed behavior.

The current experiments show an **observed difficulty**, but they do not yet explain its exact mathematical cause.

A safe conclusion is:

> **In our controlled paired-compression experiment, MeanFlow performs well at moderate compression but becomes increasingly difficult to optimize as endpoint compression becomes severe. Under the strongest tested compression, its sample-wise inverse reconstruction degrades faster than the multi-step Rectified Flow baseline.**

25 Relation to the Thesis

These experiments were designed as a simple controlled starting point before moving to medical image and segmentation-mask transport.

The real thesis problem is more difficult because an image and a segmentation mask contain different amounts of information:

$$\text{Image} \to \text{Mask}.$$

A medical image contains information such as texture, brightness, intensity, scanner characteristics and background appearance.

A segmentation mask mainly contains shape, position and class information.

Our current toy experiment does not yet reproduce complete information loss because

$$s > 0$$

remains mathematically invertible.

However, it shows that highly asymmetric endpoint geometry can already create a difficult learning problem for MeanFlow.

The next stage of the research will therefore move toward a SemFlow-style bidirectional setting and later toward the proposed medical-image system:

$$\boxed{\text{Medical Image} \leftrightarrow \text{Segmentation Mask.}}$$

Rectified Flow will be used as an important baseline, while MeanFlow will be investigated as a possible one-step or few-step alternative.

The purpose will be to observe what problems appear in the real asymmetric image-mask setting before designing the final modification.

26 Next Step

The compression experiments are now used mainly as a controlled preliminary study.

The next research stage is:

$$\boxed{\text{Rectified Flow / SemFlow-style baseline} \quad \text{vs.} \quad \text{Naive MeanFlow}}$$

for real image-to-mask and mask-to-image transport.

The initial medical experiment will avoid adding many modifications at once.

First, we will test whether a basic MeanFlow model can perform the same two directions as the Rectified Flow baseline.

After observing the real failure modes, possible modifications such as an appearance code, different uncertainty handling, improved endpoint representations, or other training changes can then be tested.

27 Conclusion

The original hypothesis was that one-step MeanFlow might immediately collapse when the destination endpoint becomes thin.

The experiments showed that this simple hypothesis was incorrect.

MeanFlow performs very well at moderate compression.

However, when compression becomes extreme, MeanFlow reconstruction becomes increasingly difficult.

Increasing the training time helps initially, but the improvement later reaches a plateau.

A direct MLP can learn the same highly compressed inverse accurately when sufficient training data are provided, showing that the inverse itself is learnable.

Finally, the Rectified Flow versus MeanFlow comparison showed that both methods degrade as compression becomes severe, but the tested shared MeanFlow model degrades more strongly and its training loss increases considerably.

These experiments do not yet establish a fundamental limitation of MeanFlow.

Instead, they provide an experimentally observed behavior and a controlled foundation for the next stage of the thesis: studying Rectified Flow and MeanFlow under the real bidirectional image-to-mask and mask-to-image problem.